Reading to Learn

Full Product Requirements Document and Tech Stack

Learn to love reading while reading to learn.

Field	Value
Document type	PRD + technical architecture
Product stage	Prototype to school pilot
Primary target	Grade 7-9 French academic reading
Extended target	Foundations down to Grade 5-6; stretch to Grade 10-11
Recommended stack	Next.js, TypeScript, Supabase Postgres, OpenAI, pgvector, Vercel
Version	v1.0

This document is written as a build-ready specification for a strong AI-assisted coder and product team. It defines the product concept, learning model, functional requirements, data model, AI pipeline, tech stack, roadmap, risks, and acceptance criteria.

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1. Executive summary

Reading to Learn is a web-first French academic reading platform for secondary students. It helps students improve reading comprehension, academic vocabulary, memory, and subject knowledge by matching each student with texts based on their interests, actual reading level, and current learning zone.

The product is intentionally not just an AI story generator. AI is used to draft, simplify, tag, and explain content. The platform itself must score difficulty, update student skill estimates, choose the next activity, schedule retrieval, trigger foundation repair, and produce auditable progress evidence.

The first real product should focus on Grade 7-9 students, because this is where many learners move from learning to read toward reading to learn. The product should also support foundation repair down to Grade 5-6 and stretch content up to Grade 10-11.

> **The central promise is: students learn to love reading while reading to learn. They enter through topics they care about, but the product routes them toward school-relevant knowledge in history, geography, science, economics, society, culture, technology, and media literacy.**

Core thesis

- Classroom teaching is one-to-many; this product gives each learner individualized practice.
- Weak readers often avoid reading because texts are too hard, too boring, or too childish.
- Teen-respectful, interest-led texts can increase reading volume.
- Controlled text difficulty can keep students in a productive 80-85% success zone.
- Spaced retrieval and cumulative review can turn reading into durable knowledge.
- Foundation repair can close prerequisite gaps before students face harder academic texts.

The non-negotiable architecture principle

> **AI generates, explains, and assists. The platform scores, adapts, remembers, repairs, and proves.**

Area	Requirement
AI responsibilities	Generate draft texts, simplify text under constraints, generate draft questions, explain vocabulary, propose tags, draft feedback, draft parent/teacher language.
Platform responsibilities	Validate AI output, score difficulty, assign levels, update skill estimates, choose next activity, schedule retrieval, trigger foundation repair, store evidence, produce reports.
Hard rule	The LLM must never be the sole authority for student level, text difficulty, next assignment, assessment claims, or progress conclusions.

2. Product definition and positioning

Working name

Reading to Learn

Tagline

Learn to love reading while reading to learn.

One-sentence description

A personalized French academic reading platform that helps secondary students improve comprehension, vocabulary, memory, and school knowledge by reading texts matched to their interests, reading level, and learning zone.

Product category

Secondary academic reading + knowledge acquisition engine.

What the product is

- A reading-to-learn platform for French academic comprehension.
- A personalized practice layer that complements classroom teaching.
- A learning-data system that tracks reading, vocabulary, knowledge, memory, and foundations.
- A tool for parents and teachers to see evidence of progress.

What the product is not

- Not just an AI story generator.
- Not a replacement for teachers.
- Not a clinical dyslexia diagnosis product.
- Not a high-stakes placement test in v1.
- Not an unrestricted AI chatbot for minors.

Positioning statements

Audience	Positioning
Primary	Personalized French academic reading for secondary students.
Student-facing	Read about what you love. Learn what school expects you to know.
Parent-facing	See what your child can read, what is still hard, and how they are improving.
Teacher-facing	Personalized reading practice, skill diagnostics, and intervention grouping for every student.
Internal	The Math Academy for academic reading and knowledge acquisition.

3. Target users and personas

Primary student segment

- Grade 7 students transitioning into denser secondary texts.
- Grade 8 students needing stronger academic vocabulary and inference.
- Grade 9 students who can decode French but struggle to learn independently from texts.

Extended student segments

- Grade 5-6 students who lack foundations before secondary school.
- Grade 10-12 students needing academic reading support, especially in expository and argumentative texts.
- Bilingual and international school students.
- French-as-second-language students.
- Francophone African students in schools where French is the language of instruction but reading foundations vary.
- Diaspora students who need to maintain academic French outside a fully French school environment.

Adult personas

Persona	Core need
Parent	Wants proof that the child is improving, not vague progress bars. Needs confidence, reading evidence, and clear next steps.
French teacher	Needs skill gaps, class grouping, assignable texts, and low-prep reading practice.
Subject teacher	Needs students to understand science, history, geography, and social studies texts.
School admin	Needs implementation simplicity, impact evidence, privacy, and teacher adoption.
Tutor / learning center	Needs diagnostics, practice between sessions, and parent-facing reports.

4. Problem statement

Student problem

Many secondary students can technically read French words, but cannot comfortably learn from dense academic texts. They struggle with long sentences, abstract vocabulary, inference, academic connectors, summarization, argument structure, disciplinary vocabulary, and reading stamina.

The result is that students can read the words but cannot reliably read to learn.

Parent problem

Parents often receive vague feedback: "read more", "needs comprehension work", or a grade with no diagnosis. They need concrete evidence: what the child can read comfortably, what still requires support, and which skills are improving.

Teacher problem

Teachers cannot personalize every text, vocabulary list, scaffold, review schedule, and foundation repair path for every student every day. In one classroom, students may differ by several grade levels in academic reading ability.

Why the problem matters

- Secondary school success depends on reading to learn across subjects.
- Weak academic reading blocks performance in history, geography, science, and exams.
- Low reading confidence reduces reading volume, which further slows growth.
- Students who need easier texts often receive childish content, which damages motivation.

5. Goals, non-goals, and scope

V1 goals

1. Diagnose a student academic French reading profile.
2. Recommend texts based on interest and ability.
3. Generate or select French texts at controlled difficulty.
4. Test comprehension, vocabulary, inference, and summary quality.
5. Keep students near an 80-85% success zone.
6. Detect foundation gaps and trigger micro-remediation.
7. Build memory through spaced retrieval and cumulative review.
8. Show parent and teacher evidence of progress.
9. Collect clean learning data for future calibration.

V1 non-goals

- Native iOS or Android apps.
- Oral fluency scoring.
- Unrestricted AI chatbot.
- Full LMS integrations.
- Full curriculum coverage for Grade 7-12.
- High-stakes placement claims.
- Clinical dyslexia diagnosis.
- Student-to-student social features.
- Open community publishing.
- Payments/subscriptions during first internal pilot.
- Complex IRT model before sufficient data exists.

Recommended v1 scope

Build a Grade 7-9 academic reading product, with foundation repair down to Grade 5-6 and stretch content up to Grade 10-11. Start with silent reading, comprehension, vocabulary, summary, retrieval, dashboards, and content review. Add oral fluency later.

6. Core learning model

The product has five learning layers: reading level, interest, knowledge transfer, foundation repair, and memory.

Layer 1: Reading level

The app estimates what kind of French text a student can handle. It should not rely on one global score only. It should track ability by text type and domain.

Text type	Description
Narrative	Stories, narrative nonfiction, personal accounts.
Biography	People, achievements, historical figures, role models.
Expository	Explanatory science, geography, society, and technology texts.
Argumentative	Claims, evidence, counterarguments, nuance, opinion texts.
Source-based	Short documents, extracts, quotes, maps/data later.

Layer 2: Interest

The app tracks both declared and revealed interests. Declared interests come from onboarding. Revealed interests come from behavior: completion rate, time on task, re-selection, success rate, and abandonment.

Cluster	Examples
Sports	football, basketball, athletics, training, injuries, tournaments
Culture	music, fashion, beauty, celebrities, food, travel
Digital life	gaming, social media, influencers, algorithms, online safety
Future/career	business, money, medicine, law, engineering, entrepreneurship
World knowledge	history, geography, politics, environment, African history
Curiosity	crime/mystery, psychology, animals, space, science

Layer 3: Knowledge transfer

Every interest should lead into school-relevant knowledge. The child chooses the door; the app chooses the curriculum hidden behind the door.

Interest	Transfer domains
Football	Geography, migration, economics, biology, statistics.
Fashion	Chemistry, trade, labor, history, sustainability.
Gaming	Psychology, logic, storytelling, probability.
Music	Poetry, physics of sound, politics, culture.
Social media	Media literacy, economics, psychology, argumentation.
Food	Chemistry, biology, geography, culture.
Crime/mystery	Law, evidence, ethics, psychology.
Cars	Physics, climate, energy, industry.

Layer 4: Foundation repair

The app detects the exact missing prerequisite that blocks progress and gives short, mature, age-respectful micro-lessons.

- Literal comprehension
- Main idea
- Inference
- Evidence selection
- Cause/consequence
- Compare/contrast
- Academic connectors
- Sentence parsing
- Pronoun/reference tracking
- Vocabulary in context
- Summarization
- Argument structure
- Disciplinary vocabulary

- Reading stamina

Layer 5: Memory

The app distinguishes storage memory from retrieval memory. Storage memory helps knowledge enter long-term memory. Retrieval memory brings knowledge back later through planned review.

Memory type	Product mechanism
Storage memory	Summaries, concept cards, vocabulary cards, repeated examples, connections to prior knowledge.
Retrieval memory	Same-day, next-day, 3-day, 7-day, 21-day, and 45-day recall questions.

7. MVP product modules

Module A: Authentication and roles

The app needs role-based authentication from day one because it serves minors and schools.

Role	Permissions summary
student	Reads, answers, receives feedback, sees own progress.
parent	Views linked child progress and reports.
teacher	Views assigned classes and student skill gaps; creates assignments.
school_admin	Manages school users, classes, and school reports.
platform_admin	Manages system data and organizations.
content_reviewer	Reviews AI-generated content without default access to student personal data.

Module B: Student onboarding

10. Student joins with school code or parent-created account.
11. Student provides grade and French background.
12. Student selects interests.
13. Student completes diagnostic.
14. App creates initial reading profile and first pathway recommendation.

Module C: Diagnostic assessment

The diagnostic should take 20-30 minutes and produce a profile, not just a single level.

Diagnostic component	Purpose
Vocabulary in context	Can the student infer meanings and recognize academic words?
Sentence comprehension	Can the student parse long or complex French sentences?
Paragraph comprehension	Can the student identify main idea, references, and implied meaning?
Expository text	Can the student read to learn from explanatory subject text?
Argumentative text	Can the student identify claims, evidence, counterclaims, and nuance?
Short summary	Can the student compress and restate key information?

Module D: Interest graph

The product should track declared strength and inferred strength per interest. Inferred strength is based on engagement, completion, and repeat selection.

```
type StudentInterest = {
  studentId: string;
  interestKey: string;
  declaredStrength: number;
  inferredStrength: number;
```

```
  sessionsCompleted: number;
  avgCompletionRate: number;
  avgSuccessRate: number;
  avgTimeOnTask: number;
  lastUsedAt: string;
};
```

Module E: Knowledge graph

The knowledge graph maps interests to school domains, concepts, prerequisites, and transfer goals. It is a key moat because it turns reading motivation into school-relevant knowledge.

```
type KnowledgeConcept = {
  id: string;
  labelFr: string;
  domain: string;
  descriptionFr: string;
  prerequisiteConceptIds: string[];
  relatedConceptIds: string[];
  gradeBandMin: number;
  gradeBandMax: number;
};
```

Module F: Text library

The text library stores approved practice texts, benchmark passages, generated drafts, versions, questions, vocabulary, skill tags, and performance data.

- Every text must have stable versions.
- Every student answer must reference text_version_id, not just text_id.
- Benchmark passages must be fixed, versioned, human-reviewed, and locked.
- Generated texts may be auto-approved only after schema validation, scoring, moderation, and factuality checks.

Module G: Reading session

A student session should take 15-25 minutes and feel like reading something interesting, not taking a test every minute.

15. Continue a pathway or choose a topic.
16. Read adaptive French text.
17. Answer comprehension questions.
18. Practice target vocabulary.
19. Write a 2-4 sentence summary.
20. Answer one retrieval question from previous knowledge.
21. Receive feedback and next recommendation.

Module H: Parent dashboard

The parent dashboard should build trust by showing evidence, confidence bands, and sample texts rather than fake precision.

Dashboard area	Content
Reading band	Grade 7.1-7.6, confidence medium.
Weekly activity	Texts completed, minutes read, questions answered.
Learning zone	Percent of sessions around 80-85% success.
Skills	Strengths and needs work.
Memory	Retrieval items reviewed and retained.
Proof texts	Comfortable, supported, and currently too difficult examples.

Module I: Teacher dashboard

The teacher dashboard should help teachers group students and intervene precisely.

View	Purpose
Class overview	Reading band, usage, completion, skill gaps.
Student detail	Progress over time, weak skills, topic engagement.
Intervention groups	Automatic grouping by weak skill or foundation gap.
Assignments	Assign text/pathway to class, group, or student.
Reports	Export class or student progress for school/parents.

Module J: Admin and content review

The admin dashboard is non-negotiable. Without content review, the app becomes a random AI generator.

- View generated text candidates.
- Approve, reject, retire, or edit content.
- See text difficulty score and feature breakdown.
- See factuality and moderation flags.
- Manage skills, vocabulary, concepts, prompt versions, and benchmarks.
- See student performance by text to recalibrate difficulty.

8. Learning engine requirements

Text difficulty engine

The app must know whether a word, sentence, question, or text is more complicated than previous ones but not too complicated for a specific student. GenAI should not be trusted alone to make this decision.

Architecture rule: GenAI writes. The scoring engine judges. Student data recalibrates.

Text features to calculate

Dimension	Features
Length	word count, paragraph count, average paragraph length
Syntax	average sentence length, max sentence length, subordinate clauses, passive forms, connectors
Vocabulary	rare words, academic words, new target words, abstract nouns, morphology
Knowledge	domain, concept novelty, prerequisite concepts, assumed background knowledge
Inference	implicit meaning, reference tracking, evidence required, multi-step reasoning
Stamina	overall density, layout, amount of uninterrupted reading
Questions	literal, vocabulary, inference, synthesis, transfer, summary difficulty

Difficulty score dimensions

```
type TextDifficulty = {
  lexical: number;      // 0-100
  syntax: number;       // 0-100
  knowledge: number;    // 0-100
  inference: number;    // 0-100
  stamina: number;      // 0-100
  question: number;     // 0-100
  overall: number;      // 0-100
  confidence: number;   // 0-100
};
```

Difficulty bands

Use internal bands rather than pretending to know exact grade precision.

Band	Description
Foundation 5A-5B	Repairing middle-primary academic reading foundations.
Foundation 6A-6B	Pre-secondary reading foundation and transition support.

Secondary 7A-7B	Primary v1 starting range for many students.
Secondary 8A-8B	More abstract, denser expository and argumentative reading.
Secondary 9A-9B	Stronger academic reading, multiple perspectives, longer texts.
Secondary 10A-10B	Stretch level for strong Grade 9 or older students.
Advanced 11-12	Later expansion, exam prep, nuanced analysis, source synthesis.

Adaptive engine

The adaptive engine should target an 80-85% success zone, enough success to preserve motivation and enough challenge to produce growth.

```
if successRate >= 0.80 && successRate <= 0.85:
  maintain level and continue pathway
elif successRate > 0.85 && successRate <= 0.95 for 2 sessions:
  increase one difficulty variable
elif successRate > 0.95:
  increase difficulty more quickly
elif successRate >= 0.70 && successRate < 0.80:
  keep level but add scaffolding
elif successRate < 0.70:
  trigger foundation repair or reduce difficulty
elif abandoned:
  shorten text or change topic
```

One-step-harder rule

The next activity should increase one or two variables at a time. Example: 450 words to 550 words, or 5 target words to 7 target words, while keeping the topic family familiar.

Skill estimate requirements

```
type StudentSkillEstimate = {
  studentId: string;
  skillId: string;
  ability: number;       // 0-100
  uncertainty: number;   // 0-100
  evidenceCount: number;
  lastEvidenceAt: string;
};
```

Foundation repair requirements

A foundation repair should be short, targeted, and age-respectful. It should not feel like primary-school remediation for older students.

Gap	Repair focus
Cause/consequence	parce que, puisque, car, donc, ainsi, par conséquent, en raison de, à cause de
Contrast	mais, pourtant, cependant, néanmoins, en revanche, alors que, malgré, bien que
Summary	Find main idea, remove details, keep cause/effect, write 2-3 sentence summary.
Inference	Find clues, distinguish stated vs implied, select evidence.
Sentence parsing	Find subject, verb, object; remove secondary clauses; rephrase simply.

Memory and retrieval requirements

Each completed reading should create concept and vocabulary review items. Retrieval should happen same session, next day, 3 days, 7 days, 21 days, and 45 days in the default v1 schedule.

```
type RetrievalCard = {
  id: string;
  studentId: string;
  conceptId: string;
  sourceTextVersionId: string;
  cardType: 'definition' | 'why_question' | 'example' | 'compare' | 'transfer' |
'short_written_response';
  promptFr: string;
  expectedAnswerFr?: string;
  rubric?: object;
  dueAt: string;
  intervalDays: number;
  easeFactor: number;
};
```

9. AI content generation and review pipeline

AI responsibilities

- Generate draft French texts under strict constraints.
- Simplify or raise text complexity under platform-specified limits.
- Generate draft questions and rubrics.
- Suggest skill, vocabulary, and knowledge tags.
- Explain vocabulary and summarize feedback.
- Draft parent/teacher report language from structured evidence.

AI must not be sole authority for

- Student reading level.
- Final text difficulty.
- Official next assignment decision.
- High-stakes assessment or placement.
- Clinical diagnosis.
- Benchmark passage creation without human review.

Generation pipeline

22. Select student profile and target learning zone.
23. Select interest, knowledge domain, concept, text type, and target skill.
24. Generate draft text using structured schema.
25. Validate schema with Zod.
26. Score text deterministically.
27. Run moderation and safety checks.
28. Run factuality checks for factual texts.
29. Generate questions and rubrics.
30. Score question difficulty and answerability.
31. Store draft candidate.
32. Auto-approve low-risk content or route to human review.
33. Assign only approved text versions to students.

Generation input schema

```
type GenerateTextInput = {
```

```
  language: 'fr';
  studentGrade: number;
  targetReadingBand: string;
  topic: string;
  primaryInterest: string;
  knowledgeDomains: string[];
  targetConcepts: string[];
  textType: 'expository' | 'argumentative' | 'biography' | 'narrative_nonfiction';
  wordCountTarget: number;
  maxAverageSentenceLength: number;
  maxNewAcademicWords: number;
  targetVocabulary: string[];
  targetSkills: string[];
  avoid: string[];
  tone: 'respectful_teen' | 'neutral_academic' | 'curious_explainer';
};
```

Generated output schema

```
type GeneratedTextCandidate = {
  title: string;
  body: string;
  estimatedReadingBand: string;
  targetVocabulary: {
    word: string;
    definitionFr: string;
    exampleSentenceFr: string;
  }[];
  knowledgeConcepts: string[];
  skillsPracticed: string[];
  questions: GeneratedQuestion[];
  safetyNotes: string[];
  factualClaims: {
    claim: string;
    confidence: 'low' | 'medium' | 'high';
    needsHumanReview: boolean;
  }[];
};
```

Question schema

```
type GeneratedQuestion = {
  questionText: string;
  questionType: 'literal' | 'vocabulary_in_context' | 'inference' | 'main_idea' |
    'cause_consequence' | 'compare_contrast' | 'evidence_selection' |
    'summary' | 'transfer';
  answerFormat: 'multiple_choice' | 'short_answer' | 'written_summary';
  choices?: string[];
  correctAnswer?: string;
  rubric?: string;
  skillIds: string[];
  difficulty: number;
};
```

Content review statuses

Status	Meaning
draft	Generated but not yet reviewed or approved.
auto_approved	Passed schema, safety, difficulty, and low-risk factual checks.
needs_human_review	Sensitive, factual, borderline, or benchmark content.
human_approved	Reviewed and approved by content reviewer.
rejected	Not assignable.
retired	Previously active but removed.
benchmark_locked	Fixed, human-reviewed benchmark passage.

10. Assessment, proof, and reporting

Assessment principle

The product should present reading level as a confidence band, not as fake exact precision. It should show evidence and examples so parents and teachers can understand the estimate.

Student profile output

```
Overall academic French reading: Grade 7.1-7.6
Confidence: medium
Narrative reading: Grade 8.0
Expository reading: Grade 6.9
Argumentative reading: Grade 6.5
Academic vocabulary: weak
Inference: medium
Summary: weak
Recommended zone: Grade 7A texts, 400-550 words
```

Parent report requirements

Report element	Requirement
Reading band	Grade range plus confidence.
Activity	Texts completed, minutes read, questions answered.
Learning zone	Share of sessions around target success zone.
Skill evidence	Strengths and needs work.
Vocabulary	Words introduced, practiced, and retained.
Retrieval	Concepts remembered over time.
Proof texts	Comfortable, supported, and too difficult examples.
Next focus	One or two concrete priorities.

Teacher report requirements

Teacher need	Requirement
Class overview	Reading bands, usage, skill gaps, low-engagement flags.
Intervention groups	Group students by weak skill or foundation gap.
Student detail	Detailed profile with evidence trail.
Assignments	Track completion and performance.
Export	PDF/CSV export later; printable summary in v1.

Benchmark passages

Benchmark passages should be fixed, versioned, human-reviewed, and periodically assigned. They allow the product to separate daily adaptive practice from more stable measurement.

- Do not dynamically generate benchmark text per student.
- Lock benchmark versions once used in assessment.

- Track benchmark results separately from practice sessions.
- Use benchmark performance to recalibrate student estimates and text difficulty.

11. Privacy, safety, and compliance

Privacy requirements

- Guardian or school consent flow for minors.
- Student-friendly privacy explanation.
- Minimal personal data collection.
- No public student profiles.
- No ads and no sale of student data.
- Data export request workflow.
- Data deletion request workflow.
- Audit logs for sensitive actions.
- Role-based access and Row Level Security from day one.

Consent record

```
type ConsentRecord = {
  studentId: string;
  guardianUserId?: string;
  consentType: 'guardian' | 'school' | 'student_over_15';
  consentVersion: string;
  privacyPolicyVersion: string;
  acceptedAt: string;
  revokedAt?: string;
};
```

AI safety requirements

- Moderate student free-text inputs.
- Moderate generated content before assignment.
- Constrain student inputs with dropdowns where possible.
- No unrestricted AI companion in v1.
- Human review for sensitive topics: politics, violence, health, sex education, religion, current events.
- No unverified statistics in generated content unless source-reviewed.
- Store prompt version and model version for each AI-generated candidate.

12. Success metrics

Student activation metrics

Metric	Definition
Onboarding completion	% of students who complete profile and interests.
Diagnostic completion	% of students who complete initial diagnostic.
First session completion	% of students who complete first reading session.
Week 1 retention	% completing 3 sessions in first week.

Engagement metrics

Metric	Definition
Weekly active students	Students completing at least one reading session weekly.
Sessions per student	Average sessions per active student per week.
Reading minutes	Average minutes read per week.
Completion rate	Completed sessions / started sessions.
Abandonment rate	Started but not completed sessions.

Interest re-selection	How often students return to chosen pathways.

Learning metrics

Metric	Definition
Learning zone rate	% of sessions around 80-85% success.
Skill growth	Change in ability estimates by skill.
Vocabulary retention	Target words retained after 7+ days.
Retrieval success	Concept retrieval accuracy over time.
Reading band movement	Movement in confidence-banded reading level.
Foundation closure	Gaps repaired and no longer blocking progression.

Trust and adoption metrics

Metric	Definition
Parent report open rate	How often parents view progress reports.
Teacher dashboard usage	Weekly teacher logins and report views.
Assignment creation	Teacher-created assignments per class.
Perceived accuracy	Parent/teacher rating of whether reports match observed ability.
Pilot retention	Schools/classes continuing after pilot period.

13. Recommended tech stack

Stack summary

```
Frontend: Next.js + TypeScript + Tailwind CSS + shadcn/ui
Backend: Next.js Server Actions / Route Handlers
Database: Supabase Postgres
Auth: Supabase Auth
Security: Supabase Row Level Security
Storage: Supabase Storage
Vector search: pgvector inside Supabase
Background jobs: Supabase Queues + Supabase Cron
Optional later: Inngest
AI: OpenAI Responses API + Structured Outputs + embeddings + moderation
Validation: Zod
Analytics: PostHog for product analytics; Postgres for learning analytics
Monitoring: Sentry
Hosting: Vercel + Supabase Cloud
Later analytics warehouse: ClickHouse or BigQuery
```

Why this stack

Choice	Reason
Next.js + TypeScript	Fast full-stack app development, strong React ecosystem, good AI-coder productivity, type safety.
Tailwind + shadcn/ui	Fast, clean dashboard UI with customizable components.
Supabase Postgres	Best fit for relational school data, learning evidence, SQL analytics, integrity, portability.
Supabase Auth + RLS	Role-based access for students, parents, teachers, and admins.
pgvector	Early semantic search and interest-to-content matching without separate vector vendor.
Supabase Queues/Cron	Background generation, scoring, moderation, retrieval scheduling, reports.
OpenAI	Controlled content generation, structured outputs, embeddings, moderation.
Zod	Validate AI and API outputs.

PostHog	Product analytics, feature flags, experiments.
Sentry	Error monitoring and performance visibility.
Vercel	Simple Next.js deployment and preview environments.

Why not Convex as primary database

Convex is attractive for a fast realtime prototype, but the long-term moat is clean learning evidence: assessments, text versions, skill estimates, reports, and calibration data. Postgres is a stronger system of record for this product.

Why not MongoDB as primary database

MongoDB can work for flexible AI metadata, but this product is a structured learning system with many relationships: student, class, teacher, text version, question, skill, answer, retrieval card, benchmark result, and report. Postgres fits that shape better.

14. Architecture and data model

High-level architecture

```
Student / Parent / Teacher / Admin Web App
        |
        v
Next.js Server Actions and Route Handlers
        |
        +--> Supabase Auth and Row Level Security
        +--> Supabase Postgres source of truth
        +--> Supabase Storage for exports/assets
        +--> Supabase Queues/Cron for background jobs
        +--> OpenAI provider interface for AI tasks
        +--> PostHog for product events
        +--> Sentry for errors and performance
```

Core database groups

Group	Tables
Identity/orgs	organizations, schools, classes, profiles, students, guardians, teacher_classes, enrollments
Learning model	skills, knowledge_domains, knowledge_concepts, student_skill_estimates, student_domain_estimates
Content	texts, text_versions, questions, question_choices, text_skills, question_skills
Vocabulary	vocabulary_items, text_vocabulary, student_word_mastery
Sessions	reading_sessions, student_answers, student_summaries, reading_session_events
Retrieval	retrieval_cards, retrieval_schedules, retrieval_attempts
AI workflow	ai_generation_jobs, ai_generated_candidates, ai_scoring_results, ai_moderation_results, prompt_versions
Reports	parent_reports, teacher_reports, benchmark_results

Core object: TextVersion

```
type TextVersion = {
  id: string;
  textId: string;
  versionNumber: number;
  title: string;
  body: string;
```

```
  language: 'fr';
  wordCount: number;
  textType: 'expository' | 'biography' | 'argumentative' | 'narrative_nonfiction' | 'source_based';
  estimatedGradeMin: number;
  estimatedGradeMax: number;
  difficulty: TextDifficulty;
  targetSkills: string[];
  targetVocabulary: string[];
  concepts: string[];
  generationType: 'human' | 'ai' | 'ai_human_reviewed';
  reviewStatus: 'draft' | 'auto_approved' | 'human_approved' | 'rejected' | 'benchmark_locked';
};
```

Core object: ReadingSessionResult

```
type ReadingSessionResult = {
  studentId: string;
  textVersionId: string;
  startedAt: string;
  completedAt?: string;
  abandoned: boolean;
  successRate: number;
  literalScore: number;
  inferenceScore: number;
  vocabularyScore: number;
  summaryScore: number;
  retrievalScore: number;
  timeOnTaskSeconds: number;
  hintsUsed: number;
  recommendedNextAction:
    | 'increase_difficulty'
    | 'maintain'
    | 'add_scaffolding'
    | 'foundation_repair'
    | 'reduce_difficulty'
    | 'change_topic';
};
```

Row Level Security access model

Role	RLS requirement
student	Can read own assignments, sessions, reports; can insert own answers; cannot read other students.
parent	Can read linked child data; cannot modify learning records.
teacher	Can read students in assigned classes and create assignments.
school_admin	Can manage school users/classes and school-level reports.
platform_admin	Full access only through service-role backend tools.
content_reviewer	Can review content; no default access to student personal data.

15. Routes, APIs, and server actions

Public and auth routes

```
/
```

```
/about
/schools
/parents
/privacy
/terms
/login
/signup
/join
/consent
/reset-password
```

Student routes

```
/student
/student/onboarding
/student/diagnostic
/student/read/[sessionId]
/student/results/[sessionId]
/student/vocabulary
/student/memory
/student/progress
/student/settings
```

Parent routes

```
/parent
/parent/students/[studentId]
/parent/reports/[reportId]
/parent/settings
/parent/privacy
```

Teacher routes

```
/teacher
/teacher/classes
/teacher/classes/[classId]
/teacher/students/[studentId]
/teacher/groups
/teacher/assignments
/teacher/reports
```

Admin routes

```
/admin
/admin/content
/admin/content/review
/admin/texts/[textId]
/admin/skills
/admin/vocabulary
/admin/concepts
/admin/ai-jobs
/admin/prompts
/admin/benchmarks
/admin/schools
```

Server actions

Area	Actions
Student	startDiagnostic, submitDiagnosticAnswer, completeDiagnostic, selectInterests, startReadingSession, submitAnswer, submitSummary, completeReadingSession, submitRetrievalAttempt
Teacher	createClass, inviteStudents, createAssignment, viewClassProgress, createInterventionGroup, exportClassReport
Parent	linkStudent, viewStudentProgress, downloadReport, requestDataExport, requestDataDeletion
Admin	createSkill, createKnowledgeConcept, generateTextCandidate, reviewTextCandidate, approveTextVersion, retireTextVersion, runDifficultyScoring, runModeration, activatePromptVersion

16. Build roadmap

Phase 0: Product skeleton

- Next.js app, Supabase project, Auth, RLS foundations.
- Basic roles: student, parent, teacher, admin.
- Dashboard shells and navigation.

Phase 1: Diagnostic and reading session

- Student onboarding and interest picker.
- Diagnostic flow with seed texts.
- Question answering, basic scoring, reading result page.

Phase 2: AI content pipeline

- AI generation jobs with structured output.
- Zod validation, text scoring engine v1, question generation.
- Moderation, factuality flagging, admin review, approved content library.

Phase 3: Adaptive engine v1

- Student skill estimates.
- Success-zone logic.
- Next-text recommendation.
- Difficulty adjustment.
- Foundation repair triggers and micro-lessons.

Phase 4: Memory system

- Retrieval cards and schedules.
- Retrieval attempts and scoring.
- Vocabulary retention and concept mastery.

Phase 5: Dashboards

- Parent weekly report.
- Teacher class dashboard.
- Intervention grouping.
- Assignments and basic exports.

Phase 6: Pilot readiness

- Benchmark passages.
- School admin tools.
- Privacy workflows and audit logs.
- Usage analytics, error monitoring, content QA workflow.

Timeline estimate with a strong AI coder

Milestone	Duration	Scope
Clickable prototype	3-6 weeks	Core flow, demo-quality.
Usable MVP	3-4 months	30-100 student test.
Pilot-ready school product	6-9 months	Class dashboards, benchmarks, privacy, content review.
Credible assessment engine	12-18 months	Calibration, validation, reliability, stronger claims.

17. QA, testing, and validation

Unit tests

- Difficulty scoring.
- Success-zone decisions.
- Retrieval scheduling.
- Skill estimate updates.
- RLS policy helper functions.
- Zod schemas.
- Question scoring.

Integration tests

- Student onboarding to diagnostic completion.
- Reading session start to completion.
- AI generation to review pipeline.
- Parent report generation.
- Teacher class report generation.

Safety tests

- Student tries unsafe prompt or input.
- Generated text contains unsafe content.
- AI outputs invalid JSON.
- AI includes unverifiable claims.
- Student tries to access another student data.
- Parent tries to access unlinked child.
- Teacher tries to access unrelated class.

Learning QA

- Manual review of text difficulty.
- Manual review of age appropriateness.
- Manual review of question answerability.
- Manual review of summary rubrics.
- Manual review of foundation repair quality.
- Manual review of retrieval prompt quality.

Validation plan

34. Start with expert-reviewed seed texts and teacher feedback.
35. Track whether app estimates match teacher perception.
36. Use benchmark passages every few weeks.
37. Compare app progress with independent teacher-administered reading tasks where available.
38. Publish internal validation report only after sufficient pilot data.

18. Risks and mitigations

Risk	Mitigation
Shallow AI story generator	Force difficulty engine, skill taxonomy, knowledge graph, retrieval, foundation repair, admin review.
Parent distrust	Use confidence bands, sample texts, skill evidence, benchmark passages, avoid fake precision.
AI factual errors	Prefer evergreen content, fact-check flags, human review for sensitive/factual content, no unverified stats.
Teacher non-adoption	Low-prep assignments, intervention grouping, exportable reports, teacher override controls.
Student low usage	Interest-led topics, teen-respectful tone, short sessions, topic choice, non-childish support.
Privacy failure	RLS from day one, role-based access, audit logs, minimal data, consent, no social features.
Over-engineering early	Start with rule-based adaptivity; upgrade to IRT/Bayesian models after data.
Bad text calibration	Combine deterministic features, expert review, benchmark passages, and student performance recalibration.

19. Appendices

Appendix A: Skill taxonomy v1

Skill key	Definition
literal_comprehension	Understand explicitly stated information.
main_idea	Identify the central idea of a paragraph or text.
inference	Infer unstated meaning from clues.
evidence_selection	Select words or sentences that justify an answer.
cause_consequence	Understand reasons and results.
compare_contrast	Understand similarity, difference, opposition, concession.
academic_connectors	Understand donc, cependant, en revanche, puisque, etc.
sentence_parsing	Identify subject, verb, object, clauses, references.
vocabulary_context	Infer word meaning from context.
summary	Condense text while preserving key meaning.
argument_structure	Identify claim, evidence, counterargument, conclusion.
disciplinary_reading	Read differently in science, history, geography, and society texts.
reading_stamina	Sustain comprehension over longer or denser texts.

Appendix B: Sample student session

Field	Example
Student	Grade 8, likes football and business, weak inference and academic vocabulary.
Recommended text	Pourquoi de jeunes footballeurs quittent leur pays.
Difficulty	Secondary 7A, 500 words, 6 target academic words.
Knowledge transfer	Geography: migration. Economics: opportunity.
Target skills	Cause/consequence, vocabulary in context, inference.
Questions	3 literal, 2 vocabulary, 2 inference, 1 summary.
Retrieval	Review concept: urbanization from previous session.
Next action	If 82% success, continue pathway with slightly longer text.

Appendix C: Sample pathway

Interest: football. Knowledge trajectory: sport -> geography -> economics -> history -> biology -> argumentation.

Step	Text idea	Knowledge transfer	Band
1	Why young footballers move countries	Migration, opportunity	7A
2	How recruiters find young talent	Networks, inequality	7B
3	Why some countries produce many players	Geography, infrastructure	8A

4	Football and colonial history	History, language, migration routes	8B
5	Training, recovery, and injury	Biology, health	8A
6	Should teenagers become professional athletes early?	Argument, ethics, education	9A

Appendix D: Database table list

Table	Purpose
organizations	Top-level account: school, tutoring center, family, internal.
schools	School profile, location, curriculum type.
classes	Class, grade, academic year.
profiles	User profile mapped to auth identity and role.
students	Student profile, grade, French background.
student_guardians	Parent/guardian link.
teacher_classes	Teacher-class permissions.
skills	Skill taxonomy.
knowledge_domains	History, geography, science, etc.
knowledge_concepts	Concept graph and prerequisites.
texts	Canonical content object.
text_versions	Versioned text body and difficulty scores.
questions	Questions linked to text versions.
question_choices	Multiple-choice options.
text_skills	Text-to-skill mapping.
question_skills	Question-to-skill mapping.
vocabulary_items	Words, definitions, examples, difficulty.
text_vocabulary	Words targeted in each text.
student_word_mastery	Vocabulary exposure and retention.
student_reading_estimates	Reading band history.
student_skill_estimates	Ability and uncertainty per skill.
student_domain_estimates	Knowledge familiarity per domain.
reading_sessions	Session-level results.
student_answers	Question-level answer evidence.
student_summaries	Student summary submissions and scores.
reading_session_events	Raw learning event log.
retrieval_cards	Memory cards.
retrieval_schedules	Due dates and intervals.
retrieval_attempts	Student recall attempts.
ai_generation_jobs	Background AI jobs.
ai_generated_candidates	Generated drafts.
ai_scoring_results	Difficulty and quality scoring results.
ai_moderation_results	Safety checks.
prompt_versions	Versioned prompts and schemas.
parent_reports	Generated parent reports.
teacher_reports	Generated teacher/class reports.
benchmark_results	Results from locked benchmark passages.

Appendix E: MVP acceptance criteria

Area	Acceptance criteria
Student	Can join, onboard, complete diagnostic, read text, answer questions, submit summary, receive feedback, complete retrieval.
Learning engine	Scores difficulty, tracks skills, tracks vocabulary, chooses next text, triggers foundation repair, schedules retrieval.
Parent	Sees reading band, weekly activity, skill strengths/weaknesses, proof texts, progress report.
Teacher	Creates class, sees students, views skill gaps, assigns reading, sees recommended groups.
Admin	Reviews generated content, approves/rejects, sees scoring/moderation, manages skills/concepts.
Safety	RLS active, consent recorded, AI outputs validated/moderated, no unrestricted chat.

Appendix F: First five things to build well

39. Diagnostic reading profile.
40. Controlled text difficulty engine.
41. Interest-to-knowledge reading paths.
42. 80-85% adaptive learning-zone logic.
43. Parent/teacher proof reports.

Final blunt recommendation: build this as a serious learning-data product, not an AI content product. AI should generate, explain, and assist. The platform should score, adapt, remember, repair, and prove.